

Antigravity Task — Phase R1: Reception Queue + Document Validation (Read)

Type: first phase of receptionist write-capable workstation build. **This phase = READ ONLY foundation.** No write actions yet. Queue viewing + document validation viewing. Writes come in R2 (route) + R3 (reject). **Effort:** ~2 days. **Scope correction:** this REVERSES Constraint #12's over-restriction. Paper documents receptionist as write-capable on both platforms; panel confirmed follow-paper + 1:1 web parity. Mobile was over-restricted → this phase begins correcting that. **Prerequisite:** web team confirmed RLS write policies ready (used in later phases; R1 is read-only so uses existing read RLS).

Context

Web team's receptionist is a full workstation (per uploaded screenshots):

- Reception Dashboard (queue counts, pending, recent triage)
- Reception Queue (5-column kanban: Submitted / AI Verified / Staff Review / Approved / Rejected)
- Document Validation (patient details, OCR text, AI confidence, view original)
- Approve & Route (triage modal → department, vitals, notes)
- Reject (reason + 20-char feedback)

Paper says receptionist is write-capable on BOTH web + mobile. Mobile currently read-only (Constraint #12 over-restriction). This build corrects mobile to match paper.

Phased approach:

- R1 (this) → Queue view + Document Validation view (READ)
 - R2 → Approve & Route (WRITE)
 - R3 → Reject w/ reason presets (WRITE)
 - R4 → Reception Dashboard
 - R5 → Constraint #12 rewrite + full tests + docs
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R1 scope → read-only foundation

Build the receptionist's ability to:

1. Navigate via bottom-nav (Dashboard placeholder / Queue / Profile)
2. See queue of submissions organized by status
3. Tap a submission → open Document Validation screen
4. View: patient details, OCR text output, AI confidence, document metadata, view original document link

NO writes this phase. Approve/Route/Reject buttons present but DISABLED w/ "Coming in next update" tooltip OR hidden entirely (agent picks — recommend disabled+labeled so the screen layout is complete for R2/R3).

Locked design

Receptionist bottom-nav

Mirror patient bottom-nav pattern for consistency:

```
[ Dashboard ] [ Queue ] [ Profile ]
```

- Dashboard → placeholder this phase (R4 builds it) → show "Dashboard coming soon" or basic welcome card
- Queue → the main R1 deliverable
- Profile → reuse existing profile pattern (receptionist's own profile, read-only staff info)

Role-gated: only `receptionist` role sees this nav. Routing dispatches receptionist → `/reception/queue` as home.

Reception Queue → tab bar layout

Mobile-native tabs (not web's 5-column kanban):

```
[ Submitted ] [ Approved ] [ Rejected ]
```

Simplification from web's 5 columns → mobile's 3 tabs:

- **Submitted** → web's "Submitted" + "AI Verified" + "Staff Review" combined (all pending

states) → receptionist sees everything needing action

- **Approved** → web's "Approved"
- **Rejected** → web's "Rejected"

Each tab → scrollable list of submission cards.

Submission card shows:

- Patient name (or "Unknown Patient" if OCR couldn't extract)
- Upload source badge (e.g., "Mobile Upload" / "Web Upload")
- Relative time ("about 17 hours ago")
- File name (e.g., patient_b_doc.pdf)
- Uploaded-by name
- Tap target → opens Document Validation

Search bar at top → filter by patient name / file name / type. (Basic client-side filter this phase; server-side search post-defense.)

Count badge per tab → number of submissions in that state.

Document Validation screen

Opened by tapping a submission card. Read-only this phase.

Sections (stacked vertically for mobile):

1. **Header** → "Document Validation" + Back to Queue + Status badge ("Pending Review" / "Approved" / "Rejected")
2. **Patient Details card:**
 - Name, Date of Birth, Gender, Contact Number, Email, Address
 - Show "—" or "Unknown Patient" for fields OCR couldn't populate
3. **OCR Text Output card:**
 - Monospace raw text of extracted document content
 - "No OCR text extraction available for this document." if empty
 - Label: "MONOSPACE RAW"
4. **AI Validation Report card:**

- Overall AI Confidence → score or "No OCR Score"
- "Confidence score not available for this upload." if none

5. Document Metadata card:

- File Name, File Type, Uploaded At, Uploaded By
- "View Original Document" button → opens document (Supabase Storage signed URL)

6. Receptionist Actions bar (bottom):

- "Reject Document" button → DISABLED this phase (R3 enables)
- "Approve & Route Patient" button → DISABLED this phase (R2 enables)
- Tooltip/label on disabled buttons: "Available soon"

What gets built

1. MODIFY `app_router.dart`

- Add receptionist route dispatch → `receptionist` role → `/reception/queue` as home
- Add routes:
 - `/reception/queue` → `ReceptionQueueScreen`
 - `/reception/document/:submissionId` → `DocumentValidationScreen`
 - `/reception/dashboard` → placeholder (R4)
 - `/reception/profile` → reuse profile
- Role guard → only receptionist reaches these
- Constraint #10 preserved → admin still blocked
- Constraint #12 → NOTE: being revised across R1-R5. This phase adds read views. Don't delete #12 yet — R5 rewrites it.

2. NEW → `reception_shell.dart` (bottom-nav scaffold)

Location: `lib/features/reception/presentation/reception_shell.dart`

- Bottom nav w/ 3 tabs (Dashboard / Queue / Profile)

- Mirror patient shell pattern
- Receptionist-only

3. NEW → `reception_queue_screen.dart`

Location: `lib/features/reception/presentation/screens/reception_queue_screen.dart`

- TabBar: Submitted / Approved / Rejected
- Each tab → ListView of submission cards
- Search bar (client-side filter)
- Count badges per tab
- Tap card → navigate to Document Validation

4. NEW → `document_validation_screen.dart`

Location:

`lib/features/reception/presentation/screens/document_validation_screen.dart`

- 5 read-only cards (patient / OCR / AI confidence / metadata)
- Disabled action bar (Reject / Approve buttons present but inert)
- View Original Document → signed URL open

5. NEW → `submission_card.dart` (widget)

Location: `lib/features/reception/presentation/widgets/submission_card.dart`

- Reusable card for queue lists

6. NEW → `reception_repository.dart`

Location: `lib/features/reception/data/reception_repository.dart`

- `getSubmissions({status})` → reads from documents/submissions table scoped by RLS
- `getSubmissionDetail(id)` → single submission w/ OCR text + AI confidence + patient details + metadata
- `getOriginalDocumentUrl(id)` → signed Storage URL
- READ ONLY this phase — no write methods

7. NEW → domain models

Location: `lib/features/reception/domain/`

- `Submission` model → `id`, `patientName`, `fileName`, `fileType`, `uploadedAt`, `uploadedBy`, `status`, `source`
- `SubmissionDetail` model → adds OCR text, AI confidence, patient details, storage path
- `SubmissionStatus` enum → `submitted`, `aiVerified`, `staffReview`, `approved`, `rejected`

Match web team's data shapes → coordinate field names w/ shared DB.

8. Reuse existing formatter helpers where applicable

- Queue status → may reuse `queue_status_formatter` pattern
 - Match existing presentation-layer helper discipline
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Data coordination w/ web team

Critical → mobile reads the SAME tables web writes.

Confirm w/ web team:

- Table name for submissions/documents (likely `documents` or `submissions`)
- Column for status, OCR text, AI confidence score
- How "Unknown Patient" is represented (null `patient_id`? unmatched?)
- Storage bucket + path pattern for original documents
- Status enum values (exact strings: `'submitted'`, `'ai_verified'`, `'staff_review'`, `'approved'`, `'rejected'`?)

Ask web team for their schema for these tables → mobile models must match exactly.
This is read-only so no migration, but field names must align.

Files affected

New:

- `lib/features/reception/presentation/reception_shell.dart`
- `lib/features/reception/presentation/screens/reception_queue_screen.dart`

- `lib/features/reception/presentation/screens/document_validation_screen.dart`
- `lib/features/reception/presentation/widgets/submission_card.dart`
- `lib/features/reception/data/reception_repository.dart`
- `lib/features/reception/domain/submission.dart`
- `lib/features/reception/domain/submission_detail.dart`
- `lib/features/reception/domain/submission_status.dart`
- `test/phase_r1_reception_queue_test.dart`
- `test/phase_r1_document_validation_test.dart`

Modified:

- `lib/core/routing/app_router.dart`
 - `files/MASTER_CONTEXT.md` (note R1-R5 in progress, #12 under revision)
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Tests

`phase_r1_reception_queue_test.dart`

Widget:

1. Receptionist login → lands on reception queue (not patient dashboard)
2. Three tabs render: Submitted / Approved / Rejected
3. Each tab shows count badge
4. Submission cards render w/ patient name, file, time
5. Tap card → navigates to Document Validation
6. Search filters list by patient name
7. Search filters by file name
8. Empty state → "No documents" per empty tab
9. Non-receptionist role → cannot reach reception queue (route guard)

`phase_r1_document_validation_test.dart`

Widget: 10. Patient details card renders all fields 11. Unknown patient → shows "Unknown Patient" + "—" fields 12. OCR text card shows extracted text 13. Empty OCR → "No OCR text extraction available" 14. AI confidence shows score 15. No score → "No OCR Score" 16. Metadata card shows file name, type, uploaded at/by 17. Reject + Approve buttons DISABLED this phase 18. View Original Document → triggers URL open (mock) 19. Back to Queue → returns to queue

Regression (must pass)

- All existing phase9 + auth + registration tests
 - Patient flow unaffected
 - Staff (department/specialist) unaffected
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Verification

```
flutter analyze
flutter test test/phase_r1_reception_queue_test.dart
flutter test test/phase_r1_document_validation_test.dart
flutter test test/phase9_registration_otp_test.dart          # regression
flutter test test/phase9_mfa_verify_test.dart               # regression
flutter test                                                 # full suite

# Constraint #1
flutter build apk --release
apktool d build/app/outputs/flutter-apk/app-release.apk -o /tmp/decompiled
grep -r "AIzaSy" /tmp/decompiled          # 0 matches
grep -r "GEMINI" /tmp/decompiled          # 0 matches
grep -r "service_role" /tmp/decompiled    # 0 matches
```

Manual test:

1. Login as receptionist test account
2. Land on Reception Queue (bottom nav visible)
3. See Submitted / Approved / Rejected tabs w/ counts
4. Tap a submission → Document Validation opens
5. Verify all 5 cards render w/ real data from shared DB
6. Verify Reject/Approve buttons disabled

7. Tap View Original → document opens
 8. Verify patient login still works (regression)
 9. Verify department/specialist staff login still works (regression)
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Screenshots (4)

- `reception_queue_submitted_tab.png` → queue w/ submitted tab active, cards visible, counts
 - `reception_queue_tabs.png` → showing all 3 tabs + search
 - `document_validation_full.png` → full validation screen w/ all cards
 - `document_validation_disabled_actions.png` → bottom action bar w/ disabled Reject/Approve buttons
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Constraint check

Constraint	Status	Notes
#1 Gemini key server-side	UNCHANGED	Re-verified
#10 Admin blocked	UNCHANGED	Admin still rejected at login
#12 Staff read-only	UNDER REVISION	R1 adds receptionist read views; #12 rewritten in R5 to reflect paper's write-capable receptionist. Do NOT delete #12 yet.
#13 Registration OTP	UNCHANGED	
#14 Session inactivity	UNCHANGED	Applies to receptionist too (10 min staff)
#15 Staff TOTP	UNCHANGED	Receptionist TOTP prompt still fires if enrolled

Defense framing (partial — full story after R5)

“The receptionist workstation on mobile mirrors the web platform per our documented one-to-one design. Phase 1 delivers the queue view and document validation screen: receptionists see patient submissions organized by status, open any submission to review extracted OCR text, AI confidence scoring, and patient details. Write actions — routing and rejection — build on this foundation in subsequent phases.”

Watch items at Gate B/D

1. Data models match web team’s shared schema exactly (field names, status enum strings) — confirm w/ web team before build
 2. Route guard → only receptionist reaches reception screens
 3. Existing staff (department/specialist) read-only views unaffected
 4. Session timeout + TOTP still apply to receptionist
 5. “Unknown Patient” edge case handled gracefully (null patient_id)
 6. Disabled action buttons clearly communicate “coming soon” not broken
 7. Signed URL for original document → expires appropriately, RLS-scoped
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Next phases (preview)

- **R2** → Approve & Route: triage modal, department dropdown (Laboratory/Imaging/Ultrasound/ECG), vitals (BP/weight/temp), triage notes, Confirm Routing → WRITES to queue. Needs RLS write policy.
- **R3** → Reject w/ reason presets: buttons (Illegible text / Unrelated files / Other), textbox, 20-char min → WRITES rejection + patient feedback.
- **R4** → Reception Dashboard: queue counts, pending submissions, recent triage activity, operational guide.
- **R5** → Constraint #12 rewrite, full integration tests, walkthrough, defense narrative.

